MILITARY SPECIFICATION

FINISH, DECK, WATER-EMULSION (FOR USE IN CONTROLLED ATMOSPHERE ENVIRONMENTS) (METRIC)

This specification is approved for use within the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. The detergent-resistant deck finish covered in this specification is intended for use in controlled atmosphere environments on all non-wood decks.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

SPECIFICATION

FEDERAL

PPP-C-96 - Cans, Metal, 28 Gage and Lighter.
PPP-F-320 - Fiberboard; Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes.

MILITARY

MIL-L-19140 - Lumber and Plywood, Fire-Retardant Treated.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 55Z3, Department of the Navy, Washington, DC 20362-5101 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 7930
DISTRIBUTION STATEMENT A Approved for public release; distribution unlimited

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T00:58Z
Check the source to verify that this is the current version before use.

STANDARDS

FEDERAL

FED-STD-313 - Material Safety Data Sheets, Preparation and the Submission of.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

2.1.2 Other Government publications. The following other Government publication forms a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DEPARTMENT OF LABOR (OSHA)

Code of Federal Regulations (CFR)

29 CFR, Part 1910.1200 - Hazard Communication Standard.

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402).

(Copies of specifications, standards, and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 1290 - Standard Test Method for Sediment in Water-Emulsion Polishes by Centrifuge.

D 1455 - Standard Test Method for 60° Specular Gloss of Emulsion Floor Polish.

D 1792 - Standard Test Method for Long-Term Removability Properties of Emulsion Floor Polishes.

D 1793 - Standard Test Method for Water Spotting of Emulsion Floor Polishes.

D 2047 - Standard Test Method for Static Coefficient of Friction of Polish-Coated Floor Surfaces as Measured by the James Machine.

D 2834 - Standard Test Method for Nonvolatile Matter (Total Solids) in Water-Emulsion Floor Polishes, Solvent-Based Floor Polishes, and Polymer-Emulsion Floor Polishes.

D 3951 - Standard Practice for Commercial Packaging. (DoD adopted)

E 70 - Standard Test Method for pH of Aqueous Solutions with the Glass Electrode. (DoD adopted)

Source: https://assist.dla.mil -- Downloaded: 202640-20T00:58Z
Check the source to verify that this is the currentversion before use.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Qualification. Deck finish furnished under this specification shall be products which are authorized by the qualifying activity for listing on the applicable qualified products list at the time set for opening of bids (see 4.3 and 6.3).

3.1.1 Standard samples. A standard sample representing the performance acceptable for use with respect to all performance factors, except toxicity and environmental, shall serve as the reference standard for qualification (see 6.4).

3.1.2 Identification of samples. The qualification sample submitted by the manufacturer shall be identified for the purpose of comparing with material subsequently submitted for acceptance. Within experimental limits, qualification and acceptance samples shall be identical. Any lot or lots found not to be identical with the qualified sample shall be rejected.

3.1.2.1 Infrared spectroscopy. The spectra of qualification and acceptance samples (see 4.3.1.1 and 4.4.3) shall be identical with respect to the number and location of infrared absorption peaks.

3.1.2.2 Color. Color, expressed as the Whiteness Index (WI), shall be measured in accordance with 4.5.12.

3.1.2.3 Resistance to soiling. The deck finish shall be resistant to soiling (see 4.5.13).

3.2 Performance. Performance shall consist of slip resistance, durability, dirt retention and general appearance in use. The product shall equal or surpass the performance of the standard sample (see 4.6).

3.3 Toxicity and environmental. The deck finish shall have no adverse effect on the health of personnel when used for its intended purpose (see 4.7). The deck finish shall contain no components which produce during normal usage noxious vapors in such concentrations as to effect the atmosphere of a controlled environment. Questions pertinent to this effect shall be referred by the contracting activity to the Naval Medical Command who will act as an advisor to the contracting activity.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T00:58Z
Check the source to verify that this is the current version before use.

3.4 Material safety data sheet. The contracting activity shall be provided a material safety data sheet (MSDS) at the time of contract award. The MSDS shall be provided in accordance with the requirements of FED-STD-313 and 29 CFR 1910.1200. When FED-STD-313 is at variance with the CFR, 29 CFR 1910.1200 shall take precedence, modify and supplement FED-STD-313. The MSDS shall be included with each shipment of the material covered by this specification (see 6.6).

3.5 Composition. Composition shall be as specified in 3.5.1 through 3.5.4.

3.5.1 Total solids. The total solids (nonvolatile content) shall be not less than 18.0 percent (see 4.5.1).

3.5.2 Sediment. Any sediment present in the deck finish shall be soft and free from grit. The amount of sediment present shall not exceed 0.2 percent by volume (see 4.5.2).

3.5.3 Alkalinity. The pH of the deck finish shall be not greater than 10.0 (see 4.5.3).

3.5.4 Stability. The finish shall develop no offensive odor, exhibit no creaming, gelling or separation, and display less than 10 percent loss of 60 degree specular gloss (see 4.5.4).

3.6 Film characteristics. Film characteristics shall be as specified in 3.6.1 through 3.6.7.

3.6.1 Removability. The dried film shall be completely removed after 75 oscillations, on a substrate of black Official Test Vinyl Asbestos Tile (OTVAT) (see 4.5.5). (Official Test Vinyl Asbestos Tile (OTVAT) is available from the Chemical Specialties Manufacturer's Association, 1001 Connecticut Ave., NW, Washington, DC 20036.)

3.6.2 Water spotting. The dried film shall show an appearance comparable to that of the standard sample, on a substrate of black OTVAT (see 4.5.6).

3.6.3 Specular gloss. The 60-degree specular gloss of the deck finish shall be equal or superior to the 60-degree specular gloss of the standard sample (see 4.5.7).

3.6.4 Scrub resistance. The dried film shall withstand 200 strokes (100 oscillations) without exposing the substrate of black OTVAT (see 4.5.8).

3.6.5 Film forming temperature. The deck finish shall form an intact film at 10 degrees Celsius (°C) (50 degrees Fahrenheit (°F)) (see 4.5.9).

3.6.6 Slip resistance. The deck finish shall have a slip resistance of 0.5 or greater (see 4.5.10 and 6.4).

3.6.7 Odor. The product shall have no offensive odor.

3.7 Unit content. The unit content shall be at least 3.785 liters (L) (1.0 gallon) of liquid at 25 ± 1°C (77 ± 2°F) (see 4.4.4).

Source: https://assist.dla.mil -- Downloaded: 20260-20T00:58Z
Check the source to verify that this is the current version before use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) Qualification inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 Qualification inspection. Qualification inspection shall be conducted at a laboratory satisfactory to the Naval Sea Systems Command (NAVSEA). Qualification inspection shall consist of the tests in 4.5 and 4.6 and the toxicological and environmental review of 4.7.

4.3.1 Qualification sampling. The samples offered for qualification by the manufacturer shall be from production batches of the same size used in producing deck finish in commercial quantities. The inspector shall take sufficient 3.785-L (1-gallon) containers for qualification inspection (see 3.1). Each container shall be labeled as follows:

DATE OF MANUFACTURE
CODE NUMBER (TO BE ENTERED BY INSPECTOR)
"KEEP FROM FREEZING" (CAN AND CONTAINER)

4.3.1.1 Certification for use and measurement of infrared spectra. In addition to the samples of 4.3.1, a 3.785-L (1-gallon) container shall be selected and forwarded for certification for use in controlled atmosphere environments and for measurement of the infrared spectra to the Officer in Charge, Annapolis Laboratory, Naval Ship Research and Development Center (Code 2833), Annapolis, MD 21402.

4.4 Quality conformance inspection. Quality conformance inspection shall consist of the examination in 4.4.4 and the tests in 4.5.

Source: https://assist.dla.mil -- Downloaded: 202610-20T00:58Z
Check the source to verify that this is the current version before use.

4.4.1 Inspection lot. A lot shall consist of all deck finish from one batch or control lot. In cases in which it is not practicable to define the sampling lot in this manner, the lot shall consist of all deck finish for delivery at one time.

4.4.2 Sampling for examination of filled containers. A random sample of filled containers shall be selected from each lot in accordance with MIL-STD-105 at inspection level S-1, and acceptable quality level (AQL) of 2.5 percent defective to verify conformance to all requirements of this specification regarding fill, closure, marking, and other requirements not involving tests.

4.4.3 Sampling for quality conformance inspection. A random sample of ten 3.785-L (1-gallon) containers shall be selected from the same batch or control lot. The containers shall be properly identified and numbered 1 to 10 inclusive. Containers 1 to 5 shall be distributed by the contracting activity or authorized representative. Of the remaining containers numbered 6 to 10, one shall be forwarded as specified in 4.3.1.1 for measurement and comparison of the infrared spectra with the qualification sample. The remaining containers shall be set aside and retained at the site of sampling pending further instructions. If the sample tested is found to not conform to any requirement of this specification, the lot represented by the sample shall be rejected.

4.4.4 Examination of filled containers. Each sample-filled container selected shall be examined for defects of the containers and the closure, for evidence of leakage, and for unsatisfactory markings. Each sample-filled container selected shall be measured to determine the volume of the contents. To determine conformance to 3.7, the liquid maintained at 25 ± 1°C (77 ± 2°F) shall be measured in a precalibrated 3.785-L (1-gallon) vessel. Any container in the sample having one or more defects, or under required fill shall be cause for rejection, and if the number of defective containers in any sample exceeds the acceptance number for the appropriate sampling plan of MIL-STD-105, this shall be cause for rejection of the lot represented by the sample.

4.4.5 Certification of quality conformance. When specified in the contract or order, a certificate of compliance shall be prepared (see 6.2.2).

4.5 Tests. The following tests shall be made on a 3.785-L (1-gallon) sample selected as specified in 4.4.3.

4.5.1 Nonvolatile matter (total solids). The nonvolatile matter of the deck finish shall be tested in accordance with ASTM D 2834 and shall meet the requirements specified in 3.5.1.

4.5.2 Sediment. The sediment of the deck finish shall be tested in accordance with ASTM D 1290 and shall meet the requirements specified in 3.5.2.

4.5.3 Alkalinity. The pH of the deck finish shall be tested in accordance with ASTM E 70 and shall meet the requirements specified in 3.5.3.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T00:58Z
Check the source to verify that this is the current version before use.

4.5.4 Stability. The stability of the deck finish shall be a period of 30 days in an oven maintained at 52 ± 1.1°C (125 ± 2°F). One hundred milliliters (mL) (3.4 ounces) of deck finish shall be placed in a 118-mL (4-ounce) glass bottle with a polyethylene-lined screwcap. The cap shall be hand tightened and the bottle and deck finish shall be placed in the oven in an inverted position. The sample shall be observed daily for the first 14 days, and thereafter twice weekly for the duration of the 30-day test period. Samples shall be examined as rapidly as possible, with the minimum of agitation. The cap shall not be loosened. The specular gloss shall be measured as specified in 4.5.7 and shall meet the requirements specified in 3.5.4.

4.5.5 Removability. The ease of removability of deck finish shall be tested in accordance with ASTM D 1792 and shall meet the requirements specified in 3.6.1.

4.5.6 Water spotting. Concurrent testing of the deck finish and the standard sample shall be tested in accordance with ASTM D 1793 and shall meet the requirements of 3.6.2.

4.5.7 Specular gloss. Concurrent testing of the deck finish and the standard sample shall be tested in accordance with ASTM D 1455, using a black glass substrate. The sample shall meet the requirements of 3.6.3.

4.5.8 Scrub resistance. Scrub resistance shall be tested in accordance with ASTM D 1792, except that the monoethanolamine (MEA) shall not be added. The requirements of 3.6.4 shall be met.

4.5.9 Film forming temperature. A sample of 0.2 mL of deck finish shall be spread over 52 square centimeters (8 square inches) on a black OTVAT substrate and refrigerated at 10 ± 0.5°C (50 ± 1°F) for 4 hours. The film shall be examined visually and by touch. The sample shall meet the requirements specified in 3.6.5.

4.5.10 Slip resistance. Slip resistance shall be tested in accordance with ASTM D 2047 and shall meet the requirements specified in 3.6.6.

4.5.11 Infrared spectra. Infrared spectra shall be tested in accordance with 4.5.11.1 and 4.5.11.2 and shall meet the requirements specified in 3.1.2.1.

4.5.11.1 Preparation of sample. A 6.3-centimeter (cm) (2.5-inch) square of aluminum foil shall be cut having a thickness of 76.2 micrometers (µm) (3 mil). One side shall be mirror bright. The square shall be washed with absolute alcohol and air dried. The mirror bright side shall give a spectra that differs less than 2 percent transmittance from the standard front surface test mirror used in the specular reflectance accessory (see 4.5.11.2). Approximately 1 mL of deck finish shall be applied with an eye dropper in a zig-zag pattern, and a uniform layer shall be produced using a 5.1-cm (2-inch) doctor blade with a 0.20-millimeter (mm) (0.008-inch) gap. The sample thus prepared shall be allowed to dry overnight before use.

4.5.11.2 Measurement of sample. The spectra of the coated foil shall be run on an infrared spectraphotometer equipped with a specular reflectance accessory. The base line of the spectra shall show 90 percent transmittance,

Source: https://assist.dla.mil -- Downloaded: 202610-20T00:58Z
Check the source to verify that this is the current version before use.

and the most intense band shall show 10 percent. It may be necessary to move the coated aluminum foil slightly to approach these conditions as nearly as possible.

4.5.12 Color. Color shall be measured in accordance with 4.5.12.1 through 4.5.12.3.

4.5.12.1 Apparatus. The apparatus used for determining the WI shall be a tristimulus colorimeter, 7.6 by 15.2 cm (3 by 6 inch) strips of Whatman No. 1 filter paper and paper clamps. The tristimulus colorimeter shall be normally furnished with an instrument standard of suitable white or light grey calibrated vitreolite. The tristimulus colorimeter shall be calibrated and operated in accordance with manufacturer's instructions.

4.5.12.2 Procedure. Duplicate samples shall be prepared by saturating the filter paper strips with deck finish and allowing to dry. This shall be done by the total immersion of the filter paper strip in the deck finish for at least 60 seconds. The saturated strip shall be removed and hung in a vertical position with a paper clamp, where it shall be allowed to dry for 24 hours at room temperature. Each sample shall be backed with five layers of plain white filter paper when dry, and four reflectance readings (tristimulus values) shall be taken on each sample.

4.5.12.3 Calculations. The WI shall be calculated from the equation:

$$WI = L - 3b.$$

The values L and b shall be measured directly with the colorimeter. The value for L shall represent reflectivity or lightness, and shall be 100 for perfect white, and 0 for perfect black. The value of b shall be related to the hue of the deck finish, and shall be positive when the hue is yellow or yellowish, negative when the hue is blue or bluish, and zero when the hue is a neutral gray. Conversion factors for colorimeters reading in X, Y, Z units are:

$$L = 10\sqrt{Y} \qquad \text{and} \qquad b = 7.0(Y - 0.847\ Z)/\sqrt{Y}.$$

4.5.13 Soil resistance. Soil resistance shall be tested in accordance with 4.5.13.1 through 4.5.13.3 (see 3.1.2.3).

4.5.13.1 Apparatus. The apparatus for soil resistance testing shall be the following: Gardner Washability Machine, Gardner Roller, Chemical Specialties Manufacturer's Association standard (CSMA) soil, white OTVAT, or equal.

4.5.13.2 Procedure. The tiles shall be cut to fit the washability machine, and the cut tiles shall be cleaned thoroughly. The sample shall be applied with cotten gauze or lamb's wool in two thin coats, allowing at least 3 hours before the application of the second coat. The tiles shall be aged for 48 hours before use. Before conducting the test, the roller shall be broken in by running it on a blank sample. Two grams of standard soil shall be spread as evenly as possible over the portion of the tiles where the roller

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T00:58Z
Check the source to verify that this is the current version before use.

will track. The machine shall be run for 300 cycles (600 passes), brushing the soil back on the track as necessary. At the end of the cycle, loose soil shall be removed by wiping the surface with a facial tissue.

4.5.13.3 Evaluation. A tile shall be prepared using the qualification sample and stored in a box so that nothing touches the soiled face. This tile shall be compared with the tile prepared using the acceptance sample. The acceptance sample shall show equal or less dirt retention than the qualification sample.

4.6 Test procedure for slip resistance and durability. The test procedure for slip resistance and durability shall be in accordance with 4.6.1 through 4.6.3.2.2 (see 3.2).

4.6.1 Selection and preparation of test deck. The test deck shall be selected and prepared in accordance with 4.6.1.1 through 4.6.1.4.

4.6.1.1 Test tiles. Light colored (such as buff or light yellow) vinyl, tile of new quality shall be selected. The deck shall be free from abnormal bumps and irregularities. Decks surrounding the test panels shall not be unreasonably dirty. The surrounding deck need not be cleaned or stripped as the test panels, but they shall be normal with respect to cleanliness.

4.6.1.2 Test conditions. Relatively heavy and as nearly equal as possible traffic conditions shall prevail on all test panels. Test panels shall not be selected in areas of abnormal traffic conditions.

4.6.1.3 Size of test panels. Each test panel shall be not less than 3.66 meters (12 feet) in length along the normal line of traffic and shall be not smaller than 5.57 square meters (60 square feet).

4.6.1.4 Preparation of test deck. The test deck shall be cleaned and thoroughly stripped of old polish by use of an effective stripper. Before application of finish, each test panel shall be rinsed three times with clear water so as to remove all traces of cleaner or detergent. The effectiveness of good cleaning shall be measured by gloss readings which shall be not greater than 10. More than one treatment may be necessary on decks which have a "built-up" film of polish.

4.6.2 Application of finish. Samples for qualification and standard samples in the correct concentration shall be applied at the same time adjacent to each other if possible, but in no case shall more than one panel be removed.

4.6.2.1 Equipment. A clean pail and a 16- to 24-ounce mop, new or like new, shall be provided for each sample. The pail and mop shall be identified with the respective code numbers of the samples to reduce the possibility of any mix-up or contamination. Mop shall be very clean, wrung almost dry of clear rinse water, and dried before it is used to apply finish.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T00:58Z
Check the source to verify that this is the current version before use.

4.6.2.2 Application. The first thin coat of each sample of finish shall be applied to the test panel which has previously been marked with its code number. Application on all panels shall be uniform. After 20 minutes of drying, the second thin coat shall be applied on top of the first as lightly as possible.

4.6.2.3 Maintenance. The test panels shall be swept daily with a soft bristle brush and damp mopped weekly with new, or like new mops and clean, cold water. Mop shall only be sufficiently damp for dirt pick-up and shall not leave water on the floor.

4.6.3 Evaluation of samples. The deck finish shall be evaluated against the standard sample for a period of 6 weeks. Evaluation shall be made immediately after application of the samples to the test deck and thereafter weekly, after maintenance but prior to traffic. Known variables from one test panel to another, such as traffic conditions or deck surfaces, shall be carefully translated to insure there is neither compromise in quality of samples, nor unfair elimination due to imposing extraordinary conditions. Another panel may be tested against the standard for confirmation. Any time the sample under test is considered inferior to the standard, it shall be rejected. For acceptance, the samples shall be equal to or better than the standard at all times for every quality considered.

4.6.3.1 Slip resistance. Relative slip resistance, compared against the standard sample, shall be determined by exerting an angular foot pressure on the deck panel using clean, dry leather soles. Any finish accumulation on the sole shall be removed. Small differences in slip resistance may be magnified by use of a clean, smooth sheet of paper under the foot.

4.6.3.2 Durability. Durability shall be determined by observing dirt retention, general appearance, gloss, and buffability.

4.6.3.2.1 Dirt retention. The degree of dirt retention, compared against the standard sample, shall be determined by visual examination, noting particularly discoloring traffic marks.

4.6.3.2.2 General appearance, gloss. The general appearance, including gloss, compared against the standard sample, shall be determined by visual examination, noting particularly nondiscoloring traffic scuff marks.

4.7 Toxicity. To determine conformance to requirements of 3.3, the manufacturer of the material shall disclose the formulation of this product to the Naval Medical Command, MEDCOM-242, Washington, DC 20372. The disclosure of proprietary information, which shall be held in confidence by the Naval Medical Command, shall include: the name, formula, and approximate percentage by weight and volume of each ingredient in the product; the results of any toxicological testing of the product; identification of its pyrolsis products; and any such other information as may be needed to permit an accurate appraisal of any toxicity problem associated with the handling, storage, application, use, disposal, or combustion of the material. Information submitted shall be clearly marked or identified to show it is being provided in connection with qualification under DOD-F-24685.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T00:58Z
Check the source to verify that this is the current version before use.

4.8 Inspection of packaging. Sample packages and packs, and the inspection of the preservation, packing and marking for shipment, stowage and storage shall be in accordance with the requirements of 4.4.2, 4.4.4, and section 5.

5. PACKAGING

(The packaging requirements specified herein apply only for direct Government acquisition.)

5.1 General.

5.1.1 Navy fire-retardant requirements.

5.1.1.1 Lumber and plywood. When specified (see 6.2.1), all lumber and plywood including laminated veneer material used in shipping container and pallet construction, members, blocking, bracing, and reinforcing shall be fire-retardant treated material conforming to MIL-L-19140 as follows:

Level A and B	-	Type II - weather resistant. Category I - general use.
Level C	-	Type I - non-weather resistant. Category I - general use.

5.1.1.2 Fiberboard. Fiberboard used in the construction of class-domestic. non-weather resistant fiberboard, and cleated fiberboard boxes shall meet the flame specified index and the specific optic density requirements of PPP-F-320 and amendments thereto.

5.2 Preservation. Preservation shall be level A, C, or commercial as specified (see 6.2.1).

5.2.1 Unit quantity. Unless otherwise specified (see 3.7 and 6.2.1), the deck finish shall be furnished in oblong 1-gallon capacity cans.

5.2.2 Level A. Cans shall conform to type V, class 4 of PPP-C-96 and the amendment thereto. Exterior coating of cans shall conform to plan B. Cans shall be provided with formed bridge type handles. Alternatively, leak-proof plastic screwcaps or leakproof plastic nozzle and screwcap closure may be used.

5.2.3 Level C. Cans shall be as specified for level A except exterior can coating shall conform to plan A.

5.2.4 Commercial. Preservation shall be in accordance with ASTM D 3951.

5.3 Packing. Packing shall be level A, B, C or commercial as specified (see 6.2.1).

5.3.1 Levels A, B and C. Six 1-gallon cans packing arrangement shall be in accordance with the exterior packing arrangement requirements of the appendix to PPP-C-96 and floor arrangement 2 by 3.

Source: https://assist.dla.mil -- Downloaded: 20260-20T00:58Z
Check the source to verify that this is the current version before use.

5.3.1.1 Level A containers. Exterior shipping containers shall be in accordance with the appendix to PPP-C-96 except that the use of fiberboard containers is prohibited.

5.3.1.2 Level B. Exterior shipping containers shall be in accordance with the appendix to PPP-C-96 except that the fiberboard containers including interior packing forms and pads shall be of the weather-resistant class. Fiberboard box closure shall be in accordance with method V of the appendix to the box specification and accomplished by use of pressure-sensitive tape.

5.3.1.3 Level C. Exterior shipping containers shall be in accordance with the level B packing requirements specified in the appendix to the can specification. Fiberboard box closure shall be in accordance with method I and accomplished by use of pressure-sensitive tape.

5.3.1.4 Commercial. Commercial packing shall be in accordance with ASTM D 3951.

5.4 Palletization. Palletization shall be in accordance with PPP-C-96.

5.5 Marking.

5.5.1 Levels A, B, and C. In addition to any special marking required (see 6.2.1) and herein, interior packs, shipping containers, and palletized loads shall be marked in accordance with PPP-C-96.

5.5.2 Commercial. In addition to any special marking required (see 6.2.1) and herein, interior packs, shipping containers, and palletized loads shall be marked in accordance with ASTM D 3951.

5.5.3 Unit containers. Unit container markings shall include:

(a) Bar coding.
(b) Manufacturer's or contractor's name or both.
(c) Lot number.
(d) Date of manufacture (month and year).

6. NOTES

6.1 Intended use. The deck finish covered in this specification is intended for use in controlled atmosphere environments on all non-wood decks on board submarines.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) When fire-retardant requirements are not required (see 5.1.1.1).
(c) Level of preservation and level of packing required (see 5.2 and 5.3).
(d) Deck finish unit quantity required (see 3.7 and 5.2.1).
(e) Special marking required (see 5.5.1 and 5.5.2).

Source: https://assist.dla.mil -- Downloaded: 202610-20T00:58Z
Check the source to verify that this is the currentversion before use.

6.2.2 Data requirements. When this specification is used in an acquisition and data are required to be delivered, the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (CDRL), incorporated into the contract. When the provisions of DoD FAR Supplement, Part 27, Sub-Part 27.475-1 (DD Form 1423) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification are cited in the following paragraph.

Paragraph no.	Data requirement title	Applicable DID no.	Option
4.4.5	Certificate of compliance	DI-E-2121	---

(Data item descriptions related to this specification, and identified in section 6 will be approved and listed as such in DoD 5010.12-L., AMSDL. Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.2.2.1 The data requirements of 6.2.2 and any task in sections 3, 4, or 5 of this specification required to be performed to meet a data requirement may be waived by the contracting/acquisition activity upon certification by the offeror that identical data were submitted by the offeror and accepted by the Government under a previous contract for identical item acquired to this specification. This does not apply to specific data which may be required for each contract regardless of whether an identical item has been supplied previously (for example, test reports).

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time set for opening of bids, qualified for inclusion in Qualified Products List QPL-24685 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligibile to be awarded contracts or purchase orders for the products covered by this specification. The activity responsible for the Qualified Products List is the Naval Sea Systems Command, SEA 55Z3, Department of the Navy, Washington, DC 20362-5101 and information pertaining to qualification of products may be obtained from that activity. Application for qualification tests shall be made in accordance with "Provisions Governing Qualification SD-6" (see 6.3.1).

6.3.1 Copies of "Provisions Governing Qualification SD-6" may be obtained upon application to Commanding Officer, Naval Publications and Forms Center, 5801 Tabor Avenue, Philadelphia, PA 19120.

6.4 Standard samples. Standard samples meeting the requirements in accordance with P-F-430 are available at cost in 3.785-L (1-gallon) cans. Inquiry should be addressed to the GSA Specification Unit - WFSIS, 7th and D St., S.W., Washington, DC 20407.

Source: https://assist.dla.mil -- Downloaded: 202610-20T00:58Z
Check the source to verify that this is the current version before use.

6.5 Slip resistance. Slip resistance is measured with the James Machine in 4.5.10 to assure that the test deck finish forms an acceptably slip-resistant surface. Slip resistance is also measured under foot on the test deck in 4.6.3.1 to assure that this slip resistance does not degrade in use. Both tests should be passed for the test deck finish to be included on the QPL.

6.6 Material safety data sheets. Contracting officers will identify those activities requiring copies of completed material safety data sheets prepared in accordance with FED-STD-313 and 29 CFR 1910.1200. The pertinent Government mailing addresses for submission of data are listed in appendix B of FED-STD-313. In order to obtain the MSDS, FAR clause 52.223-3 must be in the contract.

6.7 Subject term (key word) listing.

Alkalinity
Infrared spectroscopy
Slip resistance
Toxic products

Preparing activity:
Navy - SH
(Project 7930-N384)

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T00:58Z
Check the source to verify that this is the current version before use.

INSTRUCTIONS: In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

(Fold along this line)

(Fold along this line)

DEPARTMENT OF THE NAVY

COMMANDER
NAVAL SEA SYSTEMS COMMAND (SEA 55Z3)
DEPARTMENT OF THE NAVY
WASHINGTON, DC 20362-5101

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO. 12503 WASHINGTON D C

POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE NAVY

COMMANDER
NAVAL SEA SYSTEMS COMMAND (SEA 55Z3)
DEPARTMENT OF THE NAVY
WASHINGTON, DC 20362-5101

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T00:58Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-F-24685(SH)	FINISH, DECK, WATER-EMULSION (FOR USE IN CONTROLLED ATMOSPHERE ENVIRONMENTS) (METRIC)

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

Source: https://assist.dla.mil -- Downloaded: 202610-20T00:58Z
Check the source to verify that this is the current version before use.